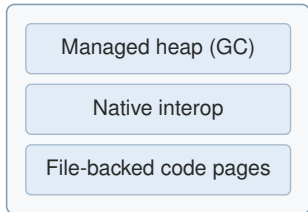


# SAGE-MM: coordinating heap, interop, and page reclamation under one embedded memory budget

## Problem



one swapless process budget

**Tensions:** heap ↔ footprint,  
allocation ↔ interop,  
residency ↔ refault. Tuned in  
isolation = globally fragile.

## SAGE-MM coordination

Heap build (**G**)  
build-time

Interop (**I**)  
source-time

static: reduce the load

**Online controller (C)**  
telemetry ⇒ pressure  
**reclamation interval  $T$**   
**+ compaction gate**  
hysteresis · cooldown · fail-closed

*only this adapts online*

## Measured: ARM32/ARM64, $n=30$

peak PSS **-24%**

GC tail **-37%**

refault **recovered**

ctrl CPU  $\approx 1\%$

## Robust across regimes

- **Favorable:** clear win
- **Neutral:** within noise (fail-safe)
- **Adverse:** bounded, guarded regression

*Every number: per-run, 95% boot-strap CI, fail-closed pipeline.*